CHAPTER 1

CONCLUDING REMARKS

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Reflect on key takeaways, highlight unique aspects of each Design Problem, and get pointers on the next steps to master System Design.

1 Conclusions

Before we conclude, we'd like to congratulate you for completing such a challenging course!
In this course, we learned system design activity from the lens of basic building blocks—the fundamental subsystems that combine to make a bigger system. Doing so enabled us to focus on the specific business use cases instead of repeating the recurring concepts.

We carefully selected our design problems from many different domains, and we've made an explicit effort to incorporate some unique aspects or issues of system design activity in each of them.

Unique Aspects of Each Design Problem

Design Problem	Unique Aspect of Design
YouTube	Building custom data stores like Vitess and BigTable to meet scalability needs
Quora	Vertical sharding of MySQL database to meet the scalability requirements
Google Maps	The use of segmentation of a map to meet scalability needs and achieve high performance
Yelp	Usage of Quadtrees for speedy access to spatial data
Uber	Improved payment service to ensure fraud detection, and matching the driver and rider on maps
Twitter	The use of client-side load balancers for multiple services that had thousands of instances in order to reduce latency
Newsfeed	A recommendation system to ensure ranking and feed suggestions
Instagram	A perfect example of how different building blocks combine to build a scalable and performant system
TinyURL	Encoding IDs in the base-58 system for generating unique short URLs
Web crawler	Detection, identification, and resolution of Web crawler traps
WhatsApp	Message management for offline users
Typeahead	The usage of an efficient trie data structure to provide suggestions
Google Docs	Concurrency management for simultaneous writes, using techniques like operational transformation (OT) and Conflict-free Replicated Data Type (CRDT)
Deployment System	Blue-green, canary, and rolling deployment strategies to achieve zero-downtime releases and minimize blast radius.
Payment System	Strong consistency and reconciliation layers to guarantee reliable, fraud-free financial transactions.
ChatGPT System	Large-scale model inference and retrieval integration using distributed GPU clusters for low-latency responses.

Table 1.1: Unique Aspects of Each Design Problem

At this point, we've built our system design toolbox, and we can now pick the right tools for the job at hand. We hope that this course has helped you gain enough knowledge to design new problems and solutions using our building blocks. Let 's put our System Design knowledge to work and build great systems!

1.1 Next steps

To expand your skill set, we recommend taking the following three courses:

- **Grokking the Generative AI System Design:** This course provides a comprehensive framework for designing scalable machine learning systems. You'll learn how to identify and handle various challenges, from data acquisition and resource estimations to model training and deployment.
- **Grokking the Principles and Practices of Advanced System Design:** Targeted for senior roles, this course elevates your system design skills by addressing complex problems such as consensus algorithms, storage systems, file systems, concurrency management, etc.

- **Grokking the Product Architecture Design Interview:** Product architecture design is essential to ensure that the services you build can effectively communicate with other systems. Mastering API design principles is critical for creating reliable and compatible APIs that support modern software architectures.

We also recommend testing your skills by taking a Mock Interviewer.
Finally, we would like to thank you for taking the time to complete this course! We hope you had a great experience. Please feel free to drop us an email or leave a comment on our community forum about any suggestions that you may have.
— Team Educative

2 Resources to Prepare for a System Design Interview

Substantial preparation is necessary to increase our odds of getting the job we apply for. Depending on a candidate's seniority and proficiency, different candidates need different times for interview preparation. For an average candidate, three to four months might be required to prepare for a System Design interview.

In the subsequent sections, we have listed additional resources to help prepare for the System Design interview in a more comprehensive way.

2.1 Additional resources

After completing this course, you have learned how to analyze, architect, and scale complex systems. While you've covered the substantial depth and breadth of traditional and fresh design problems, moving from knowledge to successful interview performance requires targeted practice.
To help you solidify your understanding and excel in your System Design interview, the following activities and resources are highly recommended to expedite your final preparation:

Free System Design lessons

- Key Concepts to Prepare for the System Design Interview
- Scalable Systems 101
- Best practices for achieving low latency in System Design
- Distributed System Design Patterns
- System Analysis vs. System Design
- API Design Interview vs. System Design Interview
- System Design fundamentals for technical product managers
- NeetCode Pro: Does it hold up for System Design interviews?
- Does Google ask System Design interview questions for L4?
- Design interview guide: Low-level OOD to Advanced System Design

- System Design Interview PDF
- The All-in-One System Design Master Template

System Design Case Studies

- How Spotify Wrapped scales for 7M users: System Design case study
- Amazon system design case study: How Amazon scales for Prime Day
- Tips from a System Design expert: How Twilio Builds for Scale
- Dropbox System Design
- Ticketmaster System Design

At Educative, we are open to sharing technical knowledge with our learners. Our comprehensive repository of blogs comprises interview prep guides, FAANG-specific insights, and in-depth technical blogs. FAANG-specific System Design interview guides include Google, Microsoft, Netflix, Amazon, and Meta.

> **Note:** You can explore Educative's blogs to find more technical articles like this one to help you ace System Design interviews.

We've also curated the most impactful technical blogs and System Design deep-dives directly from industry giants like MAANG in the following section.

Technical blogs/System Design interview articles

Many companies regularly publish the technical details of their significant work in the form of technical blogs. We can study these blogs to gain insight into the company's challenges or problems and the changes it made in the design to cope with them.

> **Quiz: Question 1:** Why are companies eager to share the technical details of their work?
> **Answer:** The reason for such sharing is to excite the technical community about the fact that the company is solving hard problems. They also hope to motivate more people to join their company. Such public blogs can also help to advertise company products to B2B customers. Additionally, such material helps the company train potential future workers on their own time. Lastly, the main reason is to let the learner know that the company has deep expertise in the specific domain or subject matter.

There's a fine line between what a company can hold back due to a competitive edge and what can be made public.
Some important technical blogs are Engineering at Meta, Meta Research, AWS Architecture Blog, Amazon Science Blog, Netflix TechBlog, Google Research, Engineering at Quora, Uber Engineering Blog, Databricks Blog, Pinterest Engineering, BlackRock Engineering, Lyft Engineering, and Salesforce Engineering.
We should always approach non-peer-reviewed(A peer-reviewed material could be a research paper that was critiqued by at least three domain expert researchers, and all points were fixed to reviewers' satisfaction before publishing in a reputable conference.) material with caution. Think about what blogs say critically and with technical acumen to decide if what they say makes sense or not. If it doesn't make sense, that could be an excellent opportunity to discuss the issue with peers.

2.2 Mock interviews

Mock interviews are a great way to prepare for System Design interviews. They involve pairing up with a friend and allowing them to ask questions. If it's not possible to use a friend, another strategy is to record ourselves and play the role of both interviewer and interviewee. With this approach, we can critique ourselves or ask a knowledgeable friend for feedback.

It's a good idea to gain experience by attending real interviews at tech companies. Once you've been through an interview or two, you'll be better able to evaluate what went right and what didn't. However, if you don't have that opportunity, you can get a feel of a real interview by attempting a personalized System Design mock interview.